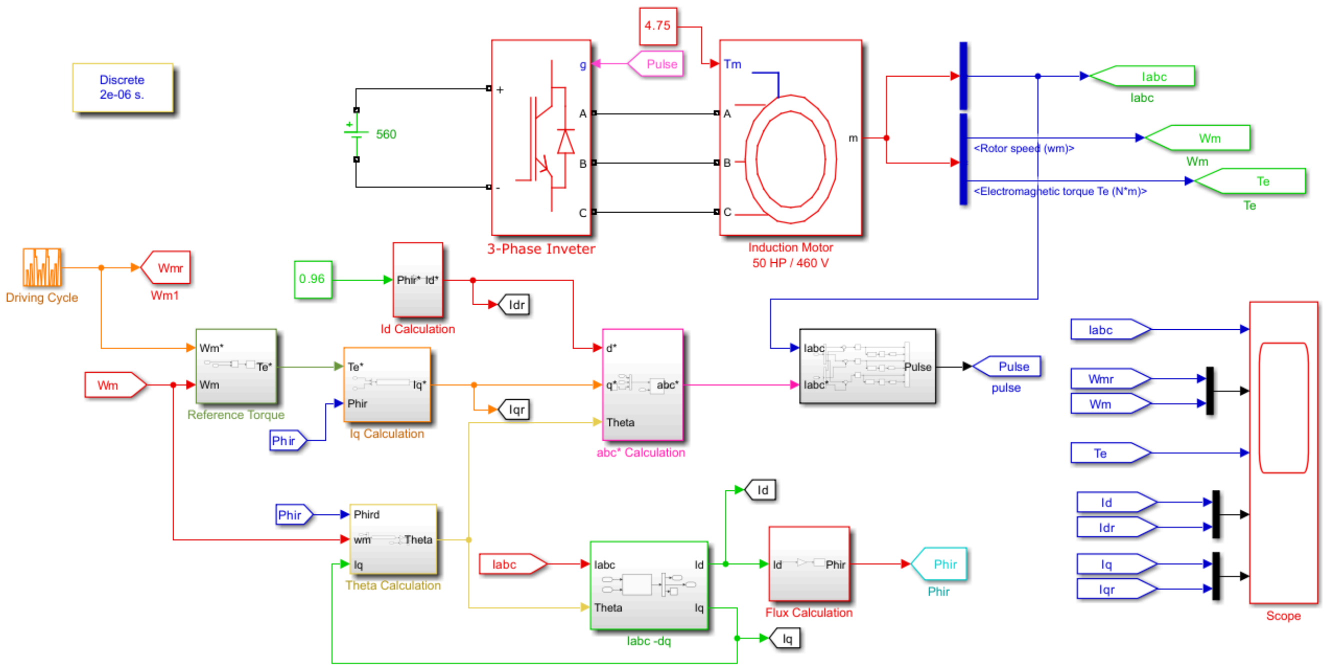
Field-Oriented Control of Induction Motor

FOC of IM with Hysteresis Current Control



Field Oriented Control of Three Phase Induction Motor - Hysteresis Current Control

Figure 6.4.1 - Simulink model for field-orientated control for induction motor using hysteresis current control.

This diagram shows the Simulink model of a field-oriented control system used for an induction motor. The model includes the three-phase inverter, the induction motor, and the control blocks used to calculate the required currents and switching signals. The speed control loop compares the reference speed with the actual motor speed to generate the torque reference. The current control loop then uses the hysteresis control subsystem to compare the actual and reference currents and generate the inverter switching pulses. The reference currents are calculated from the torque and flux control, where the d-axis current controls flux and the q-axis current controls torque.

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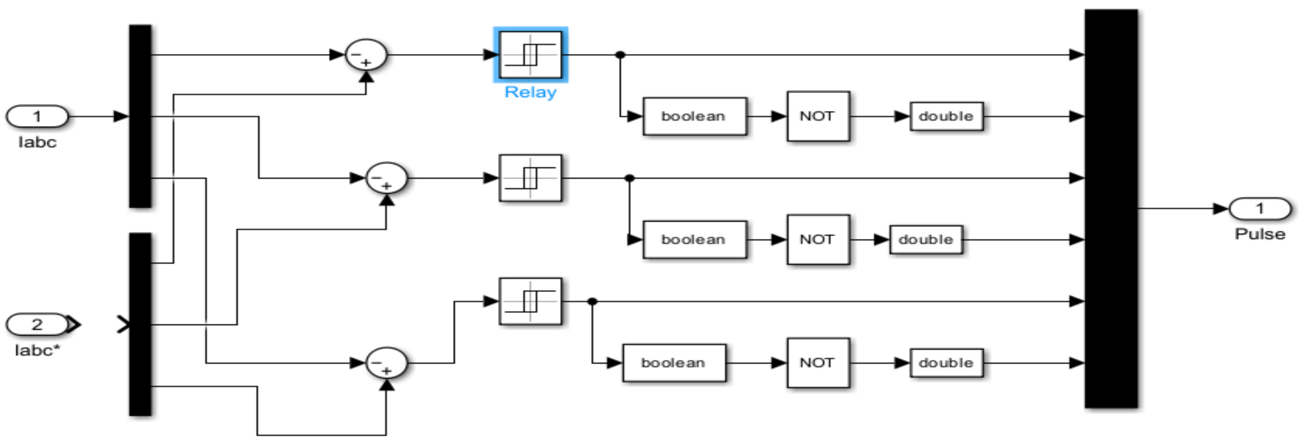

Figure 6.4.2 - Simulink model of the hysteresis current control subsystem.

This diagram shows the hysteresis current control subsystem used in the FOC model. The measured three-phase current, Iabc, is compared with the reference current, Iabc*, and the error is sent through the relay blocks for each phase. These relay blocks act as comparators and determine the switching states based on if the current error is inside or outside the hysteresis band. The logic and data conversion blocks then generate the pulse signals used to control the inverter switches.

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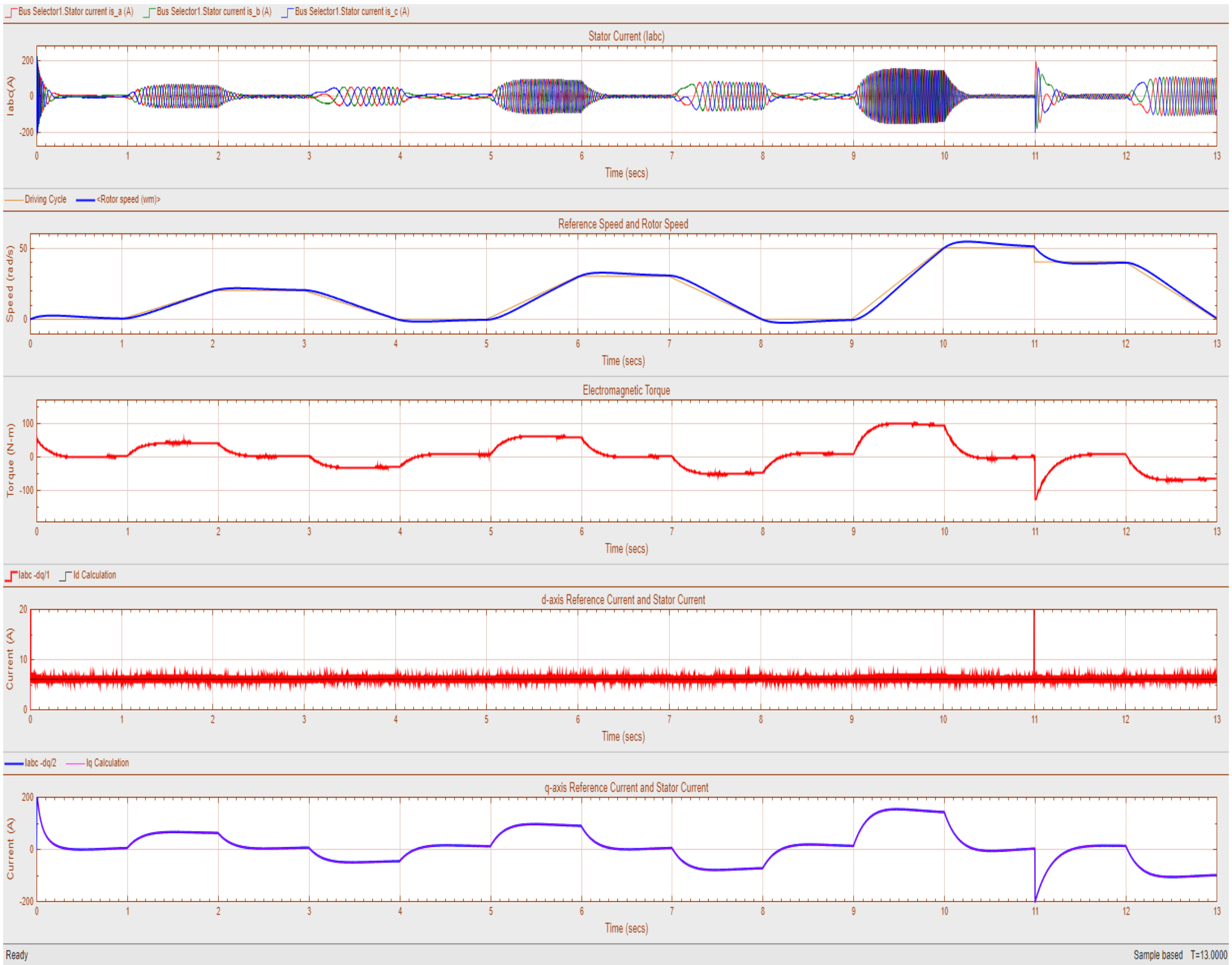

Figure 6.4.3. Simulation results of induction motor performance using field-oriented control with hysteresis current control.

As shown in the figure above, the top plot shows the three-phase stator currents, which are mostly balanced and follow a sinusoidal shape with amplitudes that change based on the operating conditions. The second plot compares the reference speed and the actual motor speed, where the motor closely follows the reference and only has a slight delay. The third plot shows the electromagnetic torque, which increases and decreases due to the hysteresis current control. The fourth plot shows the d-axis current, which is relatively constant, meaning there is proper flux control. The last plot shows the q-axis current, which changes with the torque demand and confirms that the torque control is functioning correctly.
Hysteresis current control has the advantage of being simple and having a fast response because it switches based on the current error. The main disadvantage would be that the switching frequency is not constant, which causes more ripples in current and torque, which can also cause more switching losses.

Simulink Model for FOC of IM with PI Current Control + SPWM

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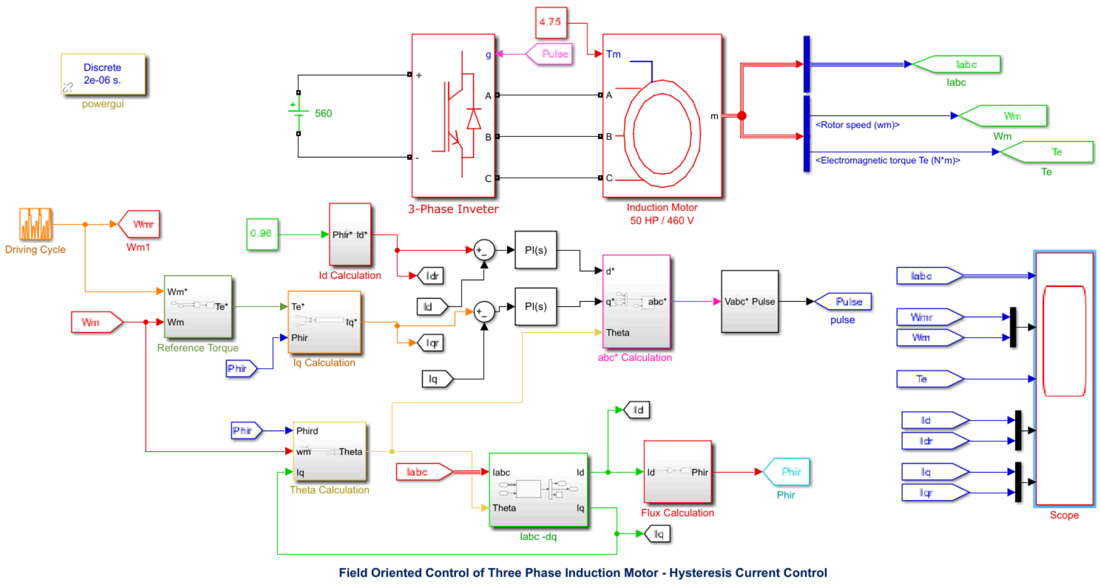

Figure 6.5.1 A Simulink model of a three-phase induction motor with PI current control + SPWM.

Figure 6.5.1 shows the full top-level diagram in Simulink of a three-phase induction motor with hysteresis control. This model is different from section 6.4 since it has the added SPWM module that had to be created which required an additional 2 PI controllers and an abc* calculation that takes the current calculation of id and iq and theta and creates a corresponding SPWM signal in order to control the motor output.

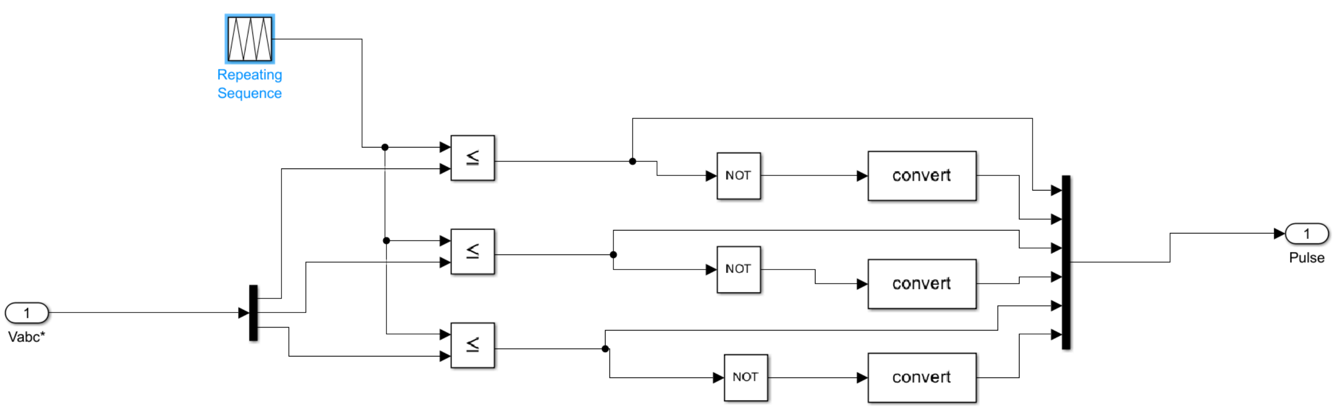

Figure 6.5.2 a Simulink submodule showcasing the Vabc* module.

Field-Oriented Control of Induction Motor

Figure 6.5.2 shows how a PWM signal is generated that goes between -1 and 1 every .5ms and is then given to the three input values, which are iq, id, and theta and alternates these signals with the NOT logical operator in order to create the SPWM signal in the hysteresis control of the motor. This pulse then goes into the three-phase inverter which controls the motor operation accordingly.

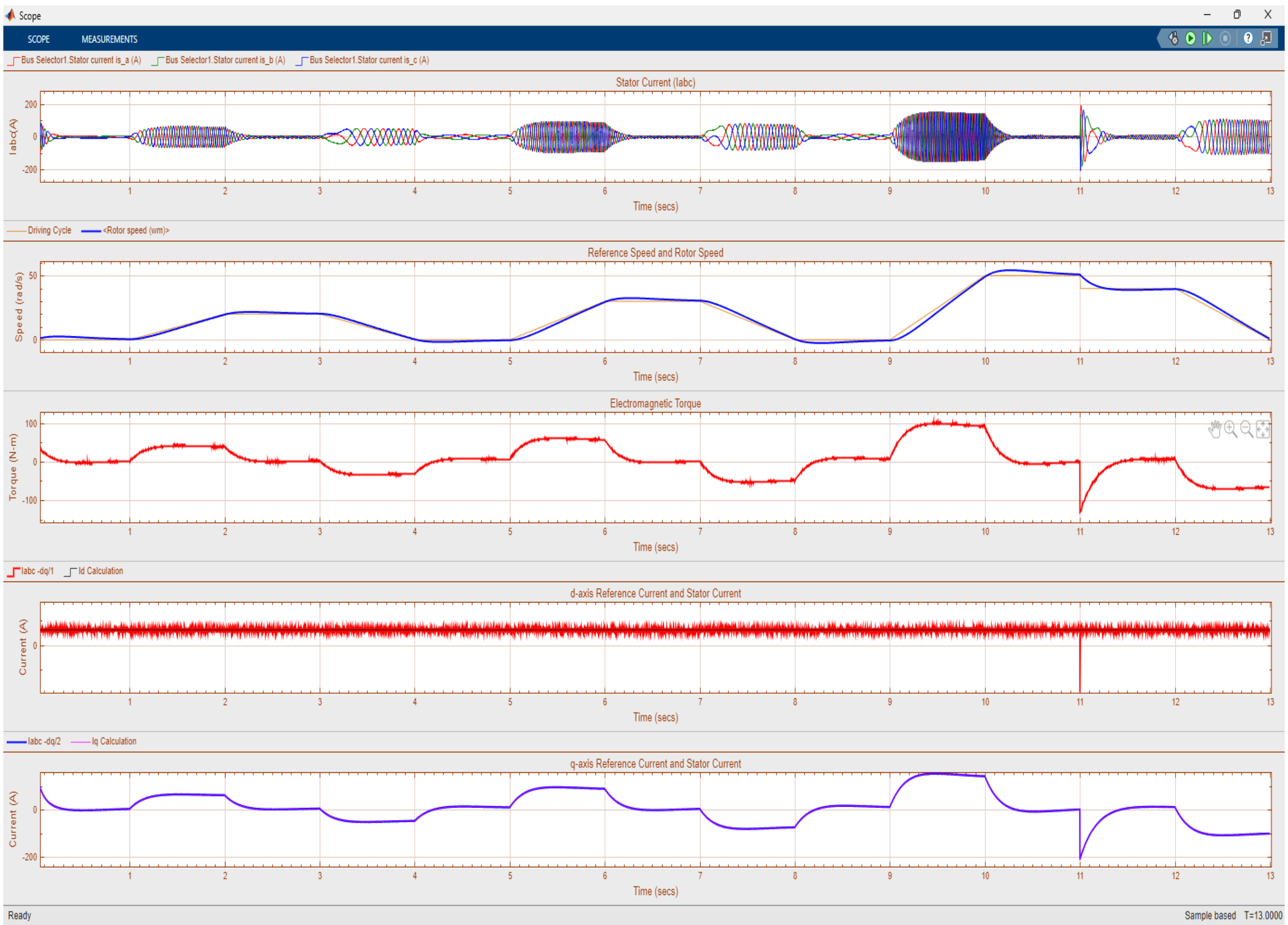

Figure 6.5.3 Corresponding Waveforms of the motor output

Figure 6.5.3 shows the corresponding waveforms of the stator currents which should all be out of phase by 120 degrees respectively and is shown with the three different colors in the waveform. The stator current is zero when the speed of the rotor is staying the same which is shown in the second waveform. The electromagnetic torque and q-axis current (waveform 3 & 5) are roughly the same waveforms with the electromagnetic waveform having half the amplitude of the q-axis reference. Both of these waveforms show a sinusoidal nature, where the values change reference, every second going from 1 to 0 to -1 to 0 back to 1, showing that the with every change the motor is going faster as the amplitude of the q-axis current increases and cause the motor to go fast. The d-axis current is fairly stable and has ripples all throughout the waveform.

The reason why PI current control with SPWM is optimal is that it creates less ripples in the electromagnetic torque, which puts less strain on the motor, having to filter these signals and allows the motor to run at a better power factor than what it was previously using in section 6.4. However, PI current control is very expensive and might not be worth it if the power company is not charging you for an excessive use of reactive power that these motors would probably create.

The current control loop[s are different from the previous model because they are subtracted from their previous values and then given to the PI controller which then creates an SPWM signal that is given directly to the three phase motor to run efficiently and use the least amount of power while obtaining full speed as quickly as possible. This feedback system is pivotal in decreasing the necessary power factor and creating a more efficient motor. We can also get a faster response out of this system since it doesn't relay on slower mechanical relays and hysteretic control to operate the motor and is instead given by a repeating waveform that can create the SPWM signals with the id,iq, and the theta position of the motor to best apply the power in the rotation sequence of the motor.